

Enhancing Time Series Forecasting through Multi-Modal Data Fusion

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Abstract

Traditional time series forecasting models typically rely on historical numerical data, rendering them vulnerable to exogenous volatility caused by real-world events. While recent multimodal approaches attempt to incorporate textual context, most rely on shallow late-fusion strategies that fail to capture the dynamic interaction between semantic cues and temporal patterns. In this work, we propose a Feature-Level Multimodal Fusion Framework that integrates foundation model embeddings into forecasting backbones via a dynamic Cross-Attention and Adaptive Gating mechanism. Unlike static fusion methods, our approach aligns semantic and numerical features in the latent space, allowing the model to selectively weigh external context against historical trends. We validate our framework on the Time-MMD benchmark across diverse domains. Experimental results demonstrate that our method significantly outperforms unimodal and late-fusion baselines in event-driven scenarios. Furthermore, we provide a critical analysis of sparse attention mechanisms, revealing that while semantic injection synergizes effectively with Global Attention (Informer), it can disrupt Locality-Sensitive Hashing (Reformer) in periodic domains. Our code is available at <https://github.com/xlesley/ese-5380-llxx>

I. INTRODUCTION

Time series forecasting serves as a cornerstone of decision-making in critical sectors such as finance, healthcare, and supply chain management. Traditional forecasting models, ranging from statistical methods like ARIMA and Prophet to classical deep neural architectures (e.g., LSTMs), have demonstrated robust performance in capturing trend-based and periodic numerical patterns. However, a significant limitation of these unimodal approaches is their exclusive reliance on historical numerical data. Consequently, they often fail to account for exogenous events that drive sudden volatility and structural breaks in real-world dynamics.

To address the complexity of temporal dependencies, recent advancements have focused on multi-scale architectures. Mechanisms such as hierarchical interpolation and pyramidal attention have allowed models to learn representations across different resolutions [1], [2]. Despite these structural improvements, these models largely focus on optimizing numerical feature extraction, often neglecting the underlying causal mechanisms behind temporal fluctuations.

Parallel to these developments, multi-modal learning has matured significantly. In domains like computer vision and natural language processing (NLP), integrating diverse modalities (text, images, audio) has proven effective for enhancing contextual understanding [3]. This success suggests a promising avenue for time series analysis: leveraging the rich semantic information embedded in textual reports or news to inform numerical predictions. While recent surveys have highlighted the potential of foundation models for time series [4], existing multi-modal forecasting approaches often rely on shallow fusion strategies (e.g., late fusion of predictions) or lack dynamic alignment between granular time series data and high-level textual semantics.

In this work, we bridge this gap by proposing a novel FEATURE-LEVEL MULTIMODAL FUSION FRAMEWORK. Unlike previous methods that rely on static statistical priors or fixed-resolution hierarchies, our approach utilizes a cross-attention mechanism to dynamically align and inject semantic context from pre-trained language models into the temporal forecasting backbone. The main contributions of this paper are summarized as follows:

- We propose a dual-stream architecture that effectively fuses numerical time series with textual embeddings derived from foundation models, enabling the model to reason about exogenous drivers of volatility.
- We introduce a dynamic Cross-Attention and Adaptive Gating mechanism that allows the model to selectively weigh semantic context against historical numerical patterns at each time step.
- We validate our approach on the Time-MMD dataset [5], demonstrating that aligned textual integration significantly improves forecasting accuracy and robustness compared to unimodal baselines.

II. RELATED WORK

A. Deep Learning for Time Series Forecasting

The evolution of deep learning has fundamentally transformed time series forecasting. Early successes with Recurrent Neural Networks (RNNs) and LSTMs have given way to Transformer-based architectures designed to capture long-range dependencies. To improve generalization and interpretability, recent research has emphasized modeling temporal features on multiple scales. For instance, N-HITS [1] introduces hierarchical interpolation and multi-rate sampling to capture signals at varying frequencies. Pyraformer [2] leverages a pyramidal attention mechanism to extract short- and long-term dependencies efficiently. Similarly, Scaleformer [6] proposes an iterative framework that decomposes forecasting tasks into multiple temporal layers. While these methods excel at numerical pattern recognition, they rely on pre-defined scale hierarchies that may not adapt well to irregular regimes caused by external events, highlighting the need for exogenous context integration.

B. Multi-modal Learning and Cross-domain Integration

Multi-modal learning aims to synthesize information from heterogeneous data sources. In the general machine learning landscape, models such as M2TR [7] and FLAVA [3] have demonstrated that shared embeddings between text and vision can be jointly optimized for superior downstream performance.

In the specific context of time series, this convergence is an emerging frontier. Jiang et al. [8] classify existing methods based on fusion strategies—early, late, and hybrid—noting the persistent challenges of temporal alignment and modality imbalance. Pioneering works like ChatTime [9] have introduced unified frameworks where language models contextualize time-dependent events. Furthermore, Chattopadhyay et al. [10] demonstrated that explicitly encoding contextual variables (e.g., weather, calendar events) enhances forecast interpretability. Our work builds upon these foundations but diverges by employing a deep feature-level fusion strategy rather than simple concatenation or late integration.

C. Foundation Models for Time Series Analysis

A new line of research investigates the applicability of Large Language Models (LLMs) and foundation models to temporal data. Kottapalli et al. [4] provide a comprehensive survey on how pretraining on diverse temporal data can yield universal representations. Kong et al. [11] argue that Multimodal Large Language Models (MLLMs) can serve as general-purpose reasoning engines to interpret time series dynamics.

While multi-modal integration has shown promise in related data quality tasks such as imputation—where linguistic context aids in reconstructing missing values—its application to direct forecasting remains under-explored. This suggests that semantic context, effectively utilized in reconstruction tasks, can similarly serve as a crucial component for robust predictive analysis.

III. DATASET

A. Dataset Overview

To evaluate the proposed framework in a realistic environment where numerical trends are heavily influenced by exogenous events, we employ the Time-MMD benchmark [5]. Existing time series datasets typically comprise only numerical records, rendering them unsuitable for verifying the efficacy of semantic reasoning. Time-MMD addresses this limitation by providing a high-quality, synchronized stream of multimodal data, aligning numerical observations with textual *facts* derived from domain-specific reports. For this study, we select diverse subsets from Time-MMD covering domains such as Economics, Public Health, and Climatology. These domains were chosen because their volatility is known to be causally linked to real-world events (e.g., policy shifts, disease outbreaks, extreme weather), thereby providing a robust testbed for multimodal fusion.

B. Data Statistics and Properties

The statistical characteristics of the selected domains are summarized in Table I. The dataset features varying granularities, from high-frequency intraday data to weekly aggregated reports, posing a challenge for the model’s adaptability.

TABLE I
HIERARCHICAL STATISTICS OF EXPERIMENTAL DATASETS. THE DATASETS ARE CATEGORIZED BY DOMAIN NATURE, DETAILING GRANULARITY, VARIABLE DIMENSION (D), TEMPORAL LENGTH (T), AND TOTAL DATA SCALE (N).

Category / Domain	Target Variable	Granularity	Dim (D)	Length (T)	Total Points (N)
I. Natural & Physical Systems					
Environment	Air Quality Index	Daily	4	11,102	44,408
Climate	Drought Level	Monthly	5	496	2,480
II. Socio-Economic Metrics					
Economy	Int'l Trade Balance	Monthly	3	423	1,269
Social Good	Unemployment Rate	Monthly	1	900	900
Security	Emergency Grants	Monthly	1	297	297
III. Sectoral Monitoring					
Energy	Gasoline Prices	Weekly	9	1,479	13,311
Health	Influenza Patients	Weekly	11	1,389	15,279
Traffic	Travel Volume	Monthly	1	531	531
Agriculture	Broiler Composite	Monthly	1	496	496

C. Data Alignment and Preprocessing

Since the raw data originates from heterogeneous sources, we apply standard preprocessing steps to ensure data quality and alignment before feeding it into the model.

- **CHRONOLOGICAL ALIGNMENT:** The Time-MMD dataset provides binary timestamps to align textual events with numerical series. We adhere to the dataset’s strict alignment protocol: for any numerical observation at time t , only the textual facts released within the interval $[t - 1, t)$ are considered valid context. This prevents look-ahead bias and ensures that the model predicts future values based solely on historically available information.
- **NUMERICAL STANDARDIZATION:** To mitigate the impact of varying scales across different dimensions, we apply Z-score normalization (Standardization) to the numerical series. Let μ_{train} and σ_{train} be the mean and standard deviation computed strictly from the training partition. The normalized value $\tilde{x}_t$ is computed as:

$$\tilde{x}_t = \frac{x_t - \mu_{train}}{\sigma_{train}} \quad (1)$$

This transformation is applied consistently to the validation and test sets to maintain distributional consistency without data leakage.

- **DATASET SPLITTING:** To strictly evaluate temporal generalization capabilities, we avoid random shuffling. Instead, the datasets are split chronologically into Training (70%), Validation (10%), and Testing (20%) subsets. This setup mimics real-world forecasting scenarios where models are trained on past data to predict unseen futures.

IV. METHODOLOGY

A. Problem Formulation

Let $\mathcal{X} = \{x_1, x_2, \dots, x_T\}$ denote a univariate or multivariate numerical time series, where $x_t \in \mathbb{R}^C$. Parallel to the numerical data, we assume access to a sequence of textual context $\mathcal{S} = \{s_1, s_2, \dots, s_T\}$, where each s_t represents the semantic information (e.g., news, reports) aligned with time step t . Given a lookback window of length L , the input consists of the numerical segment $\mathbf{X}_{in} = \{x_{t-L+1}, \dots, x_t\}$ and its corresponding textual segment $\mathbf{S}_{in} = \{s_{t-L+1}, \dots, s_t\}$. The goal of multimodal time series forecasting is to learn a function $\mathcal{F}$ that predicts the future numerical values $\mathbf{Y} = \{x_{t+1}, \dots, x_{t+H}\}$ over a horizon H :

$$\hat{\mathbf{Y}} = \mathcal{F}(\mathbf{X}_{in}, \mathbf{S}_{in}; \Theta) \quad (2)$$

where Θ represents the learnable parameters of the model.

B. Baseline Formulation

We adopt the MM-TSflib framework in Fig 1 from Time-MMD as a representative multi-modal baseline, which integrates textual prompts with numerical time-series data. The framework operates through a dual-branch architecture. The first branch employs a standard Time-Series Forecasting (TSF) backbone to capture temporal dependencies. This component is fully trainable to adapt to domain-specific patterns. The second branch processes textual context using a pre-trained BERT [12] encoder, whose parameters remain frozen to preserve semantic generalization and minimize computational cost. A trainable projection layer maps the textual embeddings (E_{text}) to the temporal feature space. The final forecast $\hat{Y}$ is obtained via a weighted additive fusion strategy:

$$\hat{Y} = (1 - \lambda) \cdot \text{TSF}(X) + \lambda \cdot (\text{Proj}(E_{text}) + Y_{prior}) \quad (3)$$

where λ serves as a balancing hyperparameter, and Y_{prior} represents the statistical prior derived from the input data.

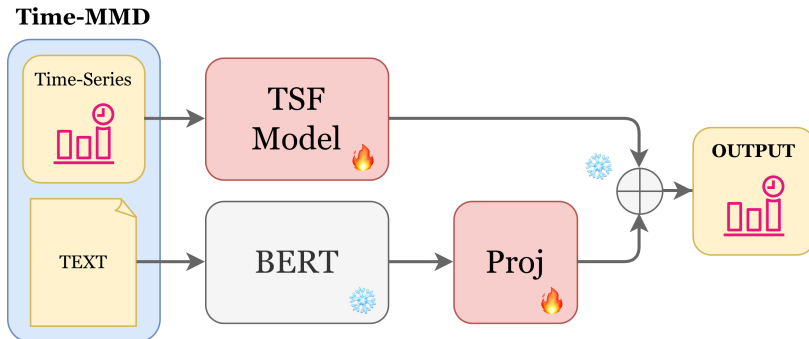


Fig. 1. Overview of the Time-MMD MM-TSflib framework Implementation.

C. Analysis of Limitations

Despite its effectiveness, we identify three critical structural constraints baseline, as illustrated in Eq.(3):

- **SHALLOW OUTPUT FUSION:** The fusion operation is performed purely on the final prediction results ($\hat{Y}$). This implies that the temporal and textual encoders operate in isolation without intermediate cross-modal attention, preventing the model from learning deep joint representations.
- **INFLEXIBLE HYPERPARAMETER:** The contribution of the textual modality is controlled by a manually tuned, global hyperparameter λ . This heuristic approach assumes a constant importance ratio between modalities across all time steps and samples, which is suboptimal for non-stationary data.
- **DEPENDENCY ON STATISTICAL PRIORS:** The text projection module relies on adding a statistical prior term (Y_{prior}) to the text embeddings to match the value range of the time series. This suggests that the text branch functions more as an offset generator for the statistical mean rather than an independent forecaster utilizing intrinsic semantic information.

D. Overview of our Framework

To address the limitations of shallow output fusion, we propose a **FEATURE-LEVEL MULTIMODAL FUSION FRAMEWORK**. As illustrated in Fig. 2, our architecture consists of a dual-stream encoder system followed by a cross-modal interaction module. The temporal branch utilizes a general Time-Series Forecasting backbone to extract latent time-series representations, while the textual branch employs a frozen pre-trained BERT encoder to capture semantic context. Unlike previous approaches that fuse modalities at the prediction head, we integrate semantic information directly into the temporal feature space via a Cross-Attention mechanism before the final projection.

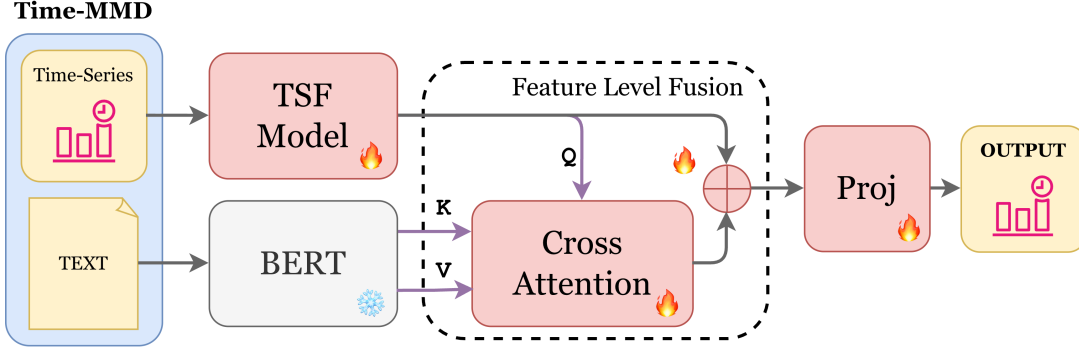


Fig. 2. Proposed Multimodal Cross-Attention Framework for Time Series Fusion

E. Cross-Modal Attention Mechanism

The core component of our proposed method is the Cross-Attention module, designed to align and inject semantic knowledge into temporal features dynamically. Let $\mathbf{H}_{ts} \in \mathbb{R}^{L \times D}$ denote the latent hidden states output by the temporal backbone, and $\mathbf{H}_{text} \in \mathbb{R}^{S \times D_{text}}$ denote the textual embeddings from the language model, where L and S represent the sequence lengths, and D and D_{text} denote the feature dimensions of the time series and text, respectively.

To model the interaction between modalities, we adopt a query-key-value attention paradigm. The temporal features serve as the Query (Q) to guide the selection of relevant semantic information, while the textual embeddings serve as both Key (K) and Value (V). The projections are defined as:

$$Q = \text{LN}(\mathbf{H}_{ts})\mathbf{W}_Q, \quad K = \text{LN}(\mathbf{H}_{text})\mathbf{W}_K, \quad V = \text{LN}(\mathbf{H}_{text})\mathbf{W}_V \quad (4)$$

where $\mathbf{W}_Q \in \mathbb{R}^{D \times D}$, and $\mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D_{text} \times D}$ are learnable projection matrices ensuring dimension alignment. $\text{LN}(\cdot)$ denotes Layer Normalization.

The context-aware semantic representations are computed via Multi-Head Attention (MHA). Crucially, we incorporate a residual connection followed by normalization and an output projection to facilitate gradient flow and feature integration:

$$\tilde{\mathbf{H}} = \text{MHA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

$$\mathbf{H}_{attn} = \text{Proj}_{out}\left(\text{LN}(Q + \tilde{\mathbf{H}})\right) \quad (6)$$

where d_k is the scaling factor corresponding to the head dimension, and Proj_{out} transforms the attended features for the subsequent gating phase.

F. Adaptive Temporal Gating

To prevent semantic noise from overwhelming the intrinsic temporal signals, we introduce a learnable Adaptive Gating Mechanism. Unlike global scalar hyperparameters, we employ a learnable parameter vector $\mathbf{G} \in \mathbb{R}^{L \times 1}$ to learn time-step-specific fusion weights, capturing the varying importance of textual context across the forecast horizon. The gate coefficient α_t at time step t is obtained via a sigmoid activation:

$$\alpha = \sigma(\mathbf{G}) \quad (7)$$

The final fused representation $\mathbf{Z}$ is computed as a weighted interpolation:

$$\mathbf{Z} = (1 - \alpha) \odot \mathbf{H}_{ts} + \alpha \odot \mathbf{H}_{attn} \quad (8)$$

$$\hat{Y} = \text{Proj}_{final}(\mathbf{Z}) \quad (9)$$

where $\odot$ denotes element-wise multiplication broadcasting across the feature dimension and Proj_{final} is a linear layer transforms the feature to final prediction. This mechanism allows the model to dynamically shift focus between historical temporal patterns ($\alpha \rightarrow 0$) and semantic contextual cues ($\alpha \rightarrow 1$) based on the learned complexity of each time step.

V. EXPERIMENTS

A. Experimental Setup

1) *Baselines*: To rigorously validate the efficacy of our proposed fusion framework, we benchmark it against a comprehensive set of established forecasting architectures. First, to assess the necessity of multi-modal integration, we compare our method with state-of-the-art *unimodal* Transformer-based models, specifically *Informer* [13] and *Reformer* [14]. These models rely exclusively on historical numerical data and serve as the primary control group to quantify the performance gain attributable to semantic context. Second, we compare against the standard MM-TSFLib implementation, which employs a late-fusion strategy where textual embeddings are linearly projected and added to the numerical forecasts. This comparison allows us to isolate the specific contribution of our proposed deep feature-level Cross-Attention mechanism versus simpler shallow fusion techniques.

2) *Implementation Details*: Our framework is implemented in PyTorch and trained on a single NVIDIA A100 GPU. Based on the temporal granularity of the *Public Health* dataset, we set the historical lookback window L to 24 time steps. The model is evaluated across four distinct prediction horizons $H \in \{12, 24, 36, 48\}$ to test its robustness over short- and long-term forecasts.

For the textual branch, we utilize a pre-trained BERT [12] model as the frozen semantic encoder. To align the modalities, we extract a text window of length 4 corresponding to the tail of the numerical lookback window and employ an ATTENTION POOLING mechanism to derive the sentence embeddings. The fusion weight is initialized at 0.1. We use the Adam optimizer for model training with an initial learning rate of $1e-4$. To ensure reproducibility, the random seed is fixed at 2021. An early stopping strategy is adopted with a patience of 10 epochs to prevent overfitting.

3) *Metrics*: We utilize Mean Squared Error (MSE) and Mean Absolute Error (MAE) as the primary evaluation metrics to quantify the deviation between predicted values and ground truth. Lower values for both metrics indicate superior forecasting accuracy.

Specifically, the Mean Squared Error (MSE) measures the average of the squares of the errors, thereby penalizing larger discrepancies more heavily than smaller ones. It is mathematically defined as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (10)$$

The Mean Absolute Error (MAE) provides the average magnitude of the errors in a set of predictions, without considering their direction, and is formulated as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (11)$$

where N represents the total number of samples in the evaluation set, y_i denotes the ground truth value for the i -th sample, and $\hat{y}_i$ corresponds to the predicted value generated by the model.

B. Main Results

To evaluate the effectiveness of the proposed Feature-Level Multimodal Fusion framework, we conducted a comparative analysis against two baselines: the pure numerical *Unimodal* models and the standard *Multimodal* baseline (MM-TSFLib from Time-MMD). Tables II and III present the quantitative results on the *Public Health* and *Traffic* datasets, respectively, using *Informer* and *Reformer* as temporal backbones.

1) *Performance on Complex Public Health Dynamics:* Table II summarizes the forecasting performance on the Public Health dataset. This domain is characterized by complex, non-stationary volatility driven by external events. The results demonstrate that incorporating semantic context generally enhances forecasting accuracy compared to the unimodal baseline.

Crucially, our proposed method significantly outperforms the standard Multimodal baseline in long-horizon scenarios. For instance, using the Informer backbone at a prediction horizon of $L = 36$, our method reduces the Mean Squared Error (MSE) from 1.568 (Standard Multimodal) to 1.345, representing a relative improvement of 14.2%. Similarly, at $L = 48$, we observe a reduction in MSE from 1.558 to 1.433. This trend suggests that while shallow fusion (Standard Multimodal) is sufficient for shorter horizons (e.g., $L = 12, 24$), deep feature-level fusion is essential for capturing the sustained impact of exogenous events over longer periods.

Results with the Reformer backbone exhibit a similar pattern, albeit with slightly higher variance. Our method achieves the lowest MSE at $L = 24$ (1.341) and $L = 36$ (1.365), confirming that the Cross-Attention mechanism effectively aligns textual semantics with sparse attention architectures.

TABLE II
PERFORMANCE COMPARISON ON PUBLIC_HEALTH DATASET

Model	Len	Unimodal		Multimodal		MM(Cross Attn)	
		MSE	MAE	MSE	MAE	MSE	MAE
Informer	12	1.036	0.680	1.022	0.672	0.965	0.671
	24	1.279	0.777	1.235	0.760	1.336	0.814
	36	1.653	0.854	1.568	0.828	1.345	0.770
	48	1.631	0.869	1.558	0.846	1.433	0.791
Reformer	12	1.159	0.752	1.096	0.743	1.392	0.816
	24	1.359	0.848	1.344	0.837	1.341	0.829
	36	1.453	0.848	1.415	0.832	1.365	0.847
	48	1.599	0.902	1.560	0.887	1.505	0.883

2) *Performance on Traffic Flows:* Table III presents the results for the Traffic dataset. Here, we observe a distinct divergence in performance based on the choice of the temporal backbone.

When paired with Informer, our Cross-Attention framework achieves state-of-the-art performance across all prediction lengths. Notably, at $L = 9$, the MSE drops dramatically to 0.194, compared to 0.304 for the Unimodal baseline and 0.277 for the Standard Multimodal approach. This substantial margin indicates that the global attention mechanism of Informer benefits significantly from the dense semantic injection provided by our gating mechanism.

Conversely, the performance with the Reformer backbone on the Traffic dataset reveals a limitation. While the Standard Multimodal baseline yields moderate improvements, our Cross-Attention method shows performance degradation in several cases (e.g., MSE increases to 0.322 at $L = 9$).

TABLE III
PERFORMANCE COMPARISON ON TRAFFIC DATASET

Model	Len	Unimodal		Multimodal		MM(Cross Attn)	
		MSE	MAE	MSE	MAE	MSE	MAE
Informer	3	0.246	0.401	0.237	0.395	0.194	0.347
	6	0.257	0.424	0.243	0.412	0.197	0.333
	9	0.304	0.474	0.277	0.446	0.194	0.336
	12	0.229	0.373	0.223	0.371	0.215	0.338
Reformer	3	0.203	0.362	0.191	0.345	0.256	0.415
	6	0.233	0.404	0.215	0.383	0.253	0.400
	9	0.202	0.364	0.193	0.354	0.322	0.471
	12	0.281	0.444	0.266	0.429	0.280	0.421

3) *Performance on Climate:* Table IV details the comparative results on the Climate dataset. This domain typically exhibits strong periodicity coupled with irregular extreme weather events, making the integration of textual warnings or reports particularly valuable.

With the Informer backbone, our proposed Cross-Attention framework demonstrates superior robustness compared to both baselines. While the Standard Multimodal approach occasionally suffers from negative transfer (e.g., increasing MSE at $L = 3$

from 0.996 to 1.169), our method consistently outperforms the Unimodal baseline. Notably, at longer horizons ($L = 6$ and $L = 9$), our method achieves significant error reductions, lowering the MSE to 1.027 and 1.056 respectively. This suggests that the proposed feature-level fusion effectively filters irrelevant noise while retaining critical semantic cues that aid in predicting climatic shifts.

In the case of the Reformer backbone, the results are more competitive. Our method maintains a slight edge at the shortest horizon ($L = 3$), but the Standard Multimodal baseline performs marginally better at extended horizons (e.g., $L = 6, 12$). This discrepancy may indicate that for highly periodic climate data, the sparse attention mechanism of Reformer aligns sufficiently well with simple concatenated features, whereas the complex gating of our Cross-Attention module yields diminishing returns when long-term dependencies are already well-captured by the model’s internal structure.

TABLE IV
PERFORMANCE COMPARISON ON CLIMATE DATASET

Model	Len	Unimodal		Multimodal		MM(Cross Attn)	
		MSE	MAE	MSE	MAE	MSE	MAE
Informer	3	0.996	0.809	1.169	0.857	0.984	0.803
	6	1.266	0.894	1.308	0.913	1.027	0.808
	9	1.279	0.885	1.261	0.893	1.056	0.819
	12	1.288	0.894	1.191	0.872	1.125	0.846
Reformer	3	1.025	0.818	1.023	0.818	1.021	0.804
	6	1.134	0.851	0.970	0.784	1.012	0.804
	9	1.150	0.857	1.037	0.815	1.055	0.821
	12	1.046	0.822	1.040	0.819	1.091	0.839

4) *Performance on Economy*: The forecasting results for the Economy dataset are presented in Table V. Economic data is notoriously challenging due to its high volatility and susceptibility to rapid market shifts, resulting in higher performance variance across different models.

For the Informer backbone, we observe a “mid-horizon sweet spot.” Our method significantly outperforms baselines at medium prediction lengths ($L = 6$ and $L = 9$), reducing the MSE at $L = 6$ from 1.043 (Unimodal) to 0.861. This indicates that textual data (e.g., market news) provides valid leading indicators for medium-term trends. However, at the longest horizon ($L = 12$), the Unimodal model regains superiority. This phenomenon suggests that at extended ranges, the semantic signals may become decorrelated with numerical trends, introducing noise that outweighs the benefits of fusion.

The Reformer backbone exhibits a distinct anomaly at $L = 6$, where our method achieves a remarkable performance breakthrough, reducing MSE from 1.431 (Unimodal) to 0.685—a relative improvement of over 52%. This specific case highlights the potential of our Cross-Attention mechanism to capture critical turning points in economic time series that standard methods miss. However, similar to the Informer case, performance degrades at $L = 12$, reinforcing the observation that aligning high-dimensional semantic features with long-term economic forecasting remains a non-trivial challenge requiring careful calibration of the fusion horizon.

TABLE V
PERFORMANCE COMPARISON ON ECONOMY DATASET

Model	Len	Unimodal		Multimodal		MM(Cross Attn)	
		MSE	MAE	MSE	MAE	MSE	MAE
Informer	3	0.558	0.697	0.634	0.770	0.652	0.775
	6	1.043	0.978	0.870	0.905	0.861	0.895
	9	1.158	1.035	1.008	0.976	0.904	0.923
	12	0.777	0.848	0.864	0.903	1.298	1.113
Reformer	3	0.636	0.674	0.600	0.659	0.876	0.878
	6	1.431	1.120	1.171	1.015	0.685	0.773
	9	0.362	0.497	0.298	0.453	0.973	0.938
	12	0.385	0.528	0.337	0.495	1.294	1.104

C. Analysis of the Informer and Reformer

This section investigates the theoretical underpinnings of the performance divergence observed between the Informer and Reformer backbones. Our experiments indicate that while the proposed Feature-Level Fusion framework exhibits strong synergy with Informer, it frequently induces negative transfer when applied to Reformer, particularly in datasets with strong periodicity like Traffic. We attribute this phenomenon to the fundamental conflict between *semantic injection* and *locality-sensitive hashing*.

1) *Synergy with ProbSparse Attention (Informer)*: The core mechanism of Informer is ProbSparse Attention, which optimizes efficiency by selecting a subset of “Active Query” based on the Kullback-Leibler (KL) divergence between the query’s attention distribution and a uniform distribution.

Our analysis suggests that the injection of textual semantics serves as a Enhancer. In standard time series, the numerical values during the onset of an event (e.g., a sudden traffic jam) might not immediately exhibit a statistical distribution distinct enough to be selected as an Active Query. However, the fused representation $\mathbf{Z}_t$ incorporates the high-dimensional embedding of the event report. This semantic injection pushes the feature vector $\mathbf{Z}_t$ toward the heavy-tail region of the feature distribution, effectively increasing its distinctiveness. Consequently, the ProbSparse mechanism is more likely to select these event-driven time steps for global attention computation, allowing the model to focus its capacity on the most volatile—and textually correlated—periods.

2) *Manifold Distortion in LSH Attention (Reformer)*: Conversely, Reformer relies on Locality Sensitive Hashing (LSH) to approximate full attention. LSH operates on the premise that queries q_i and keys k_j that are close in the high-dimensional embedding space should be hashed into the same bucket. This assumption holds well for unimodal periodic data, where historical patterns (e.g., “8:00 AM yesterday”) are vectorially similar to current patterns (“8:00 AM today”).

However, our Feature-Level Fusion introduces a phenomenon we term *Semantic Manifold Distortion*. Let $\mathbf{h}_t$ be the numerical temporal feature and $\mathbf{e}_t$ be the textual embedding. The fused vector is

$$\mathbf{z}_t = (1 - \alpha)\mathbf{h}_t + \alpha \cdot \mathbf{e}_t \quad (12)$$

- When an exogenous event occurs at time t , the term $\alpha \cdot \mathbf{e}_t$ acts as a significant displacement vector in the feature space.
- Even if the underlying numerical state $\mathbf{h}_t$ is similar to a historical state $\mathbf{h}_{t-k}$ (periodicity), the presence of the semantic vector ensures that $\mathbf{z}_t$ is far distinct from $\mathbf{z}_{t-k}$ (assuming no event occurred at $t - k$).

This displacement causes the LSH mechanism to assign $\mathbf{z}_t$ and $\mathbf{z}_{t-k}$ to different hash buckets, effectively breaking the model’s ability to retrieve relevant historical contexts based on similarity. In this scenario, valid semantic context acts as noise to the hashing algorithm, degrading the retrieval rate of periodic patterns and leading to the observed performance drop in the Traffic domain.

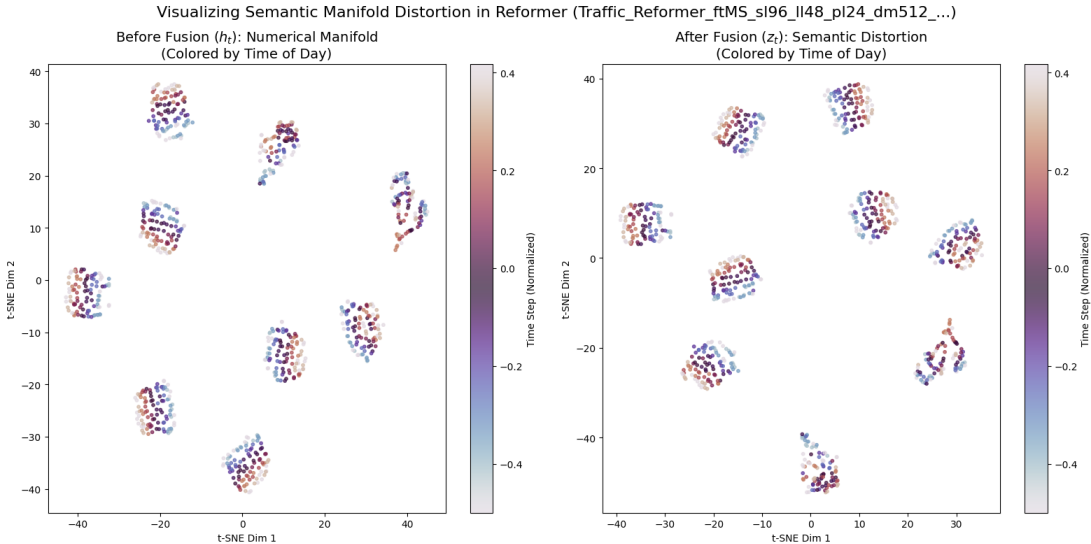


Fig. 3. **Visualizing Semantic Manifold Distortion.** Comparison of the latent feature space before (Left) and after (Right) multimodal fusion using t-SNE. The color gradient represents the Time of Day. The injection of semantic vectors introduces spatial shifts and dispersion to the numerical manifold. This distortion disrupts the locality-sensitive hashing (LSH) mechanism in Reformer, as historically similar points (which should be hashed together) are displaced into different buckets due to semantic noise.

3) *Granularity Mismatch Hypothesis*: A secondary factor is the frequency mismatch between modalities. Traffic and Climate data are characterized by high-frequency periodic signals, which Reformer excels at capturing via local clustering. Textual facts, however, are sparse and low-frequency. Injecting sparse semantic discontinuities into a high-frequency continuous manifold disrupts the local smoothness required for effective LSH clustering. In contrast, Informer’s global view (via representative

queries) is robust to such local discontinuities, allowing it to treat semantic signals as global trend indicators rather than local noise.

VI. CONCLUSION

A. Summary

In this work, we addressed the limitation of unimodal time series forecasting models in capturing exogenous volatility driven by real-world events. We proposed a FEATURE-LEVEL MULTIMODAL FUSION FRAMEWORK that integrates textual semantics from foundation models into temporal backbones via a dynamic Cross-Attention and Adaptive Gating mechanism. Unlike traditional shallow fusion strategies that combine modalities only at the output stage, our approach aligns numerical and textual features in the latent space, allowing the model to weigh semantic context dynamically. Extensive experiments on the Time-MMD benchmark demonstrated that our framework significantly improves forecasting accuracy in event-driven domains such as Public Health and Economics, outperforming both unimodal baselines and standard late-fusion multimodal methods.

B. Insights

Our comparative analysis reveals critical interactions between multimodal fusion strategies and sparse attention mechanisms:

- **SYNERGY WITH GLOBAL ATTENTION:** We find that semantic injection acts as a “relevance amplifier” for ProbSparse attention (Informer). By shifting event-aligned time steps toward the heavy-tail of the feature distribution, the fused representations become more distinguishable, enabling the attention mechanism to better capture critical turning points.
- **CONFLICT WITH LOCALITY-SENSITIVE HASHING (LSH):** Conversely, we observe a phenomenon of *Semantic Manifold Distortion* when applying deep fusion to LSH-based architectures (Reformer). The high-dimensional displacement caused by semantic vectors disrupts the local clustering required by LSH, leading to negative transfer in highly periodic domains like Traffic. This suggests that locality in numerical time series does not necessarily translate to locality in the fused semantic-numerical space.
- **THE “MID-HORIZON” SWEET SPOT:** In volatile domains like Economy, textual signals serve as effective leading indicators for medium-term horizons ($L = 6, 9$). However, their predictive power decays at extended horizons, suggesting a temporal bound on the validity of current news for future forecasting.

C. Limitations

Despite the promising results, our framework faces several limitations:

- **BACKBONE SENSITIVITY:** As evidenced by the performance degradation on the Traffic dataset with Reformer, our method is not universally compatible with all efficient attention mechanisms. The reliance on feature-level alignment requires the temporal backbone to be robust to manifold shifts induced by semantic embeddings.
- **GRANULARITY MISMATCH:** The current approach aligns text and numerical data using a fixed window. In high-frequency domains (e.g., intraday traffic), textual events are sparse compared to numerical sampling rates. Injecting sparse semantic signals into dense continuous manifolds can introduce noise, as the model may struggle to distinguish between a lack of news and neutral news.
- **DEPENDENCE ON ALIGNMENT QUALITY:** The model’s performance is strictly bound by the chronological accuracy of the text-numerical alignment. It assumes that relevant news appears exactly within the lookback window, whereas in reality, market reactions to news may have variable lags or lead times not captured by rigid windowing.

D. Future Work

Future extensions may address these limitations along several directions. Domain-adaptive or parameter-efficient tuning of the text encoder could improve semantic grounding and reduce representation mismatch. Multi-granular alignment mechanisms, incorporating both high-frequency and low-frequency textual signals, may better capture the heterogeneous temporal impact of semantic events. Hybrid fusion strategies that combine early, late, and cross-attention pathways may offer more flexible control over when semantic information should influence temporal representations. Integrating diverse text sources—such as social media signals, structured metadata, or expert summaries—would enable a richer evaluation of multimodal forecasting under realistic conditions. Finally, incorporating intervention-aware or causal modeling could allow the framework to explicitly reason about policy shocks, exogenous events, and counterfactual scenarios, expanding its applicability to decision-making contexts.

APPENDIX A

HYPERPARAMETERS

TABLE VI
HYPERPARAMETERS USED ACROSS ALL EXPERIMENTS.

Category	Parameter	Value
Time-Series Model	Model Type	Informer, Reformer
	Sequence Length (L)	36
	Label Length	18
	Prediction Horizon	3, 6, 9, 12
	Hidden Dimension (d)	512
	FFN Dimension (d_{ff})	2048
	Attention Heads	8
	Encoder Layers	2
Training	Batch Size	32
	Learning Rate	1×10^{-4}
	Optimizer	Adam
	Epochs	10
	Early Stopping	Patience = 5
Text Encoder	Encoder	BERT (frozen)
	Text Window Size	4
	Embedding Dimension	768
	Projection Dimension	512
	Pooling Method	Average
Fusion Module	Gating Function	Sigmoid Gate
	Output Combination	Residual Sum
General Settings	Random Seed	2021
	Num Workers	10
	Frequency	Monthly (m)

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